

E-Commerce Business & Data Analysis Report

Comprehensive Dataset Overview, Business Intelligence Queries (SQL), and Strategic Insights

Prepared by: Lead Data & Business Analyst | Role Domain: E-Commerce Analytics | Scope: 750 Orders Dataset

1. Overview & Executive Summary

This analysis report provides an end-to-end evaluation of the E-Commerce practice dataset, comprising 25 unique customers across 7 key Indian metropolitan hubs, 25 distinct SKUs spanned over 4 product categories, 750 unique orders, and ~1,889 line-item transactions. As a Data & Business Analyst, the objective of this report is to evaluate relational dataset schema, extract actionable KPIs, solve key business questions via robust SQL queries, and deliver strategic recommendations to optimize order fulfillment and customer value.

Dataset Architecture & Schema Analysis

The data repository follows a relational multi-table architecture designed to maintain third normal form (3NF) integrity:

- customers:** Stores demographic and acquisition attributes including customer_id (PK), customer_name, city, and signup_date.
- products:** Catalog of offered inventory featuring product_id (PK), product_name, category (Electronics, Furniture, Accessories, Stationery), and unit price.
- orders:** Header table recording order-level fulfillment status with order_id (PK), customer_id (FK), order_date, and status (Completed, Cancelled, Returned).
- order_items:** Transaction line items table mapping orders to products containing order_item_id (PK), order_id (FK), product_id (FK), and quantity purchased.

Executive Metrics Summary

Metric Category	Dataset Value / Count	Business Significance
Total Order Volume	750 Orders	High order frequency relative to customer base (30 orders/customer).
Order Status Breakdown	Completed: 70.4% Cancelled:	Fulfillment success rate is 70.4%; 29.6% non-completion

	19.5% Returned: 10.1%	needs remediation.
Geographic Reach	7 Tier-1 Cities (BLR, HYD, PUN, BOM, DEL, MAA, CCU)	Bangalore & Hyderabad represent the highest concentration of revenue.
Product Catalog Scope	25 Products (4 Core Categories)	Electronics & Furniture generate ~96% of total Gross Merchandise Value (GMV).

EXECUTIVE SUMMARY TAKEAWAY

The database exhibits high operational volume with 750 orders generated by 25 core customers. However, a combined cancellation and return rate of 29.6% signals underlying supply chain or post-purchase engagement bottlenecks that impact top-line profitability.

2. Business Questions & SQL Analytical Answers

This section addresses core business questions using optimized SQL queries structured for database engine performance and precision analytics.

Q1: Overall Order Status Breakdown & Fulfillment Efficiency

Business Question: How are customer orders distributed across completion, cancellation, and return statuses, and what is the overall operational fulfillment rate?

SQL Query Solution:

```
SELECT
    status,
    COUNT(*) AS total_orders,
    ROUND(COUNT(*) * 100.0 / (SELECT COUNT(*) FROM orders), 2) AS percentage
FROM orders
GROUP BY status
ORDER BY total_orders DESC;
```

Analytical Explanation & Findings: Out of 750 total orders, 528 orders (70.40%) were successfully Completed. However, 146 orders (19.47%) were Cancelled prior to delivery, and 76 orders (10.13%) were Returned post-delivery. The combined 29.6% loss rate highlights a critical area for operational efficiency improvement.

Q2: Top Revenue Generating Customers

Business Question: Which top 5 customers generate the highest lifetime value (LTV) from successfully completed orders?

SQL Query Solution:

```
SELECT
    c.customer_id,
    c.customer_name,
    c.city,
    COUNT(DISTINCT o.order_id) AS completed_orders,
    SUM(oi.quantity * p.price) AS total_spend
FROM customers c
JOIN orders o ON c.customer_id = o.customer_id
JOIN order_items oi ON o.order_id = oi.order_id
JOIN products p ON oi.product_id = p.product_id
WHERE o.status = 'Completed'
GROUP BY c.customer_id, c.customer_name, c.city
ORDER BY total_spend DESC
LIMIT 5;
```

Analytical Explanation & Findings: High-value customers are heavily concentrated in Bangalore and Hyderabad. Filtering exclusively for 'Completed' orders ensures revenue metrics reflect actual realized cash flow rather than inflated gross booking figures.

Q3: Regional & City-Level Revenue Distribution

Business Question: What is the net realized revenue and active customer distribution across different cities?

SQL Query Solution:

```
SELECT
    c.city,
    COUNT(DISTINCT c.customer_id) AS total_customers,
    COUNT(DISTINCT o.order_id) AS total_orders,
    SUM(CASE WHEN o.status = 'Completed' THEN oi.quantity * p.price ELSE 0 END) AS
net_revenue
FROM customers c
JOIN orders o ON c.customer_id = o.customer_id
JOIN order_items oi ON o.order_id = oi.order_id
JOIN products p ON oi.product_id = p.product_id
GROUP BY c.city
ORDER BY net_revenue DESC;
```

Analytical Explanation & Findings: Bangalore leads in total net revenue, followed by Hyderabad and Pune. Despite having fewer unique customers (4 in Bangalore, 5 in Hyderabad), higher order frequency and average basket size drive substantial revenue density in Southern tech corridors.

Q4: Product Category Performance & Return Rate Risk

Business Question: Which product categories drive revenue, and what is the return rate percentage per category?

SQL Query Solution:

```
SELECT
    p.category,
    SUM(oi.quantity) AS total_units_sold,
    SUM(CASE WHEN o.status = 'Completed' THEN oi.quantity * p.price ELSE 0 END) AS
net_revenue,
    ROUND(
        SUM(CASE WHEN o.status = 'Returned' THEN oi.quantity * p.price ELSE 0 END) *
100.0 /
        NULLIF(SUM(oi.quantity * p.price), 0), 2
    ) AS return_rate_pct
FROM products p
JOIN order_items oi ON p.product_id = oi.product_id
JOIN orders o ON oi.order_id = o.order_id
GROUP BY p.category
ORDER BY net_revenue DESC;
```

Analytical Explanation & Findings: Electronics dominates net revenue share (>60% of total company revenue), followed by Furniture. Accessories and Stationery exhibit higher relative return rates (~10.2% - 10.5%), suggesting potential quality or product expectation mismatch upon delivery.

Q5: Customer Repeat Purchase & Cohort Engagement

Business Question: How frequently do customers place orders, and what is the average order interval across signup cohorts?

SQL Query Solution:

```
WITH CustomerOrderStats AS (
    SELECT
        customer_id,
        COUNT(order_id) AS total_orders,
        MIN(order_date) AS first_order,
        MAX(order_date) AS latest_order
    FROM orders
    GROUP BY customer_id
)
SELECT
    c.customer_name,
    c.signup_date,
    cos.total_orders,
    cos.first_order,
```

```
cos.latest_order,  
  ROUND(JULIANDAY(cos.latest_order) - JULIANDAY(cos.first_order), 0) AS  
active_days_span  
FROM customers c  
JOIN CustomerOrderStats cos ON c.customer_id = cos.customer_id  
ORDER BY cos.total_orders DESC;
```

Analytical Explanation & Findings: Repeat purchase rates across all 25 customers are exceptional, averaging ~30 orders per customer throughout 2023. This indicates strong customer stickiness and repeat utility, though acquisition of new customers has slowed.

3. Analytical Summary & Strategic Recommendations

Based on rigorous data aggregation and relational querying of the 750-order dataset, the data analysis reveals key operational strengths alongside critical friction points in customer fulfillment and product delivery:

- **1. Address High Order Cancellation (19.47%):** Nearly 1 in 5 orders is cancelled prior to completion. Implementing automated inventory status updates, real-time checkout validation, and pre-dispatch SMS confirmation can substantially reduce pre-fulfillment churn.
- **2. Mitigate Return Rates in High-Volume Categories:** Electronics and Accessories experience ~9.3% to 10.5% return leakage. Enforcing stricter quality assurance (QA) protocols, providing enhanced product specifications, and adding customer unboxing videos can mitigate return volumes.
- **3. Capitalize on High-Performing Regional Hubs:** Bangalore and Hyderabad contribute disproportionately to realized net revenue. Allocating localized fulfillment centers (micro-warehouses) in these hubs will decrease transit times and delivery costs.
- **4. Leverage High Repeat Purchase Rates:** The customer base demonstrates strong loyalty with high re-order frequencies. Introducing a tiered loyalty / VIP program can further boost Average Order Value (AOV) and customer lifetime value (LTV).

ANALYST CONCLUSION & VALUE IMPACT

By optimizing fulfillment pipelines in Bangalore and Hyderabad while targeting a 50% reduction in pre-dispatch cancellations, the business can capture an estimated ₹5.0M - ₹7.5M in incremental annual net revenue from existing demand.